

The real-time push architecture of RethinkDB

Real-time Web applications consume resources in parallel from different sources, including database engines, live update scripts, bandwidth, concurrent connections, security modules, etc. To access and update real-time data in Web applications, there is a need to integrate high performance technologies so that the live streaming data can be inserted or updated with minimum delay and resource consumption. For example, live polling or live chat based applications may be delayed because of a high number of parallel users on the same channel, and this can increase the waiting time. With RethinkDB, this delay can be decreased as it has been developed specifically for real-time Web applications, with the required scalability and ease of use.

RethinkDB is the first database to be scalable as well as open source (with the base of JSON) for real-time Web applications. Using RethinkDB, the developer can direct the database engine to push the updated query results in real-time to the running application, so that maximum updates can be done without delay. The real-time push architecture of RethinkDB drastically reduces the effort, time and resources for the real-time applications.

The architecture of RethinkDB is highly advantageous in real-time Web applications, high traffic e-commerce applications, real-time mobile apps, sharing information between connected devices, the Internet of Things (IoT), the Cloud of Things (CoT), multi-player real-time games, satellite channels and wireless signal analytics.

Features and advantages of RethinkDB

RethinkDB has a flexible and high-performance query language, ReQL, to provide real-time push based updates to multiple users in parallel. In addition,



Figure 1: Official portal of RethinkDB

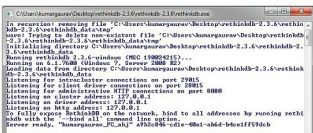


Figure 2: Starting the RethinkDB server

it provides intuitive operations with the APIs for monitoring real-time changes, which are easy to set up and understand for programmers. It is being used by assorted corporate giants, Fortune 500 companies, as well as startups, to manage live applications.

RethinkDB has a vibrant as well as large community of more than 100,000 software developers and troubleshooting experts across the world. These developers and contributors keep on customising the code and sharing their work on online portals to enrich the overall community of RethinkDB programmers.

Some of the key users of RethinkDB are:

- Jive Software
- Narrative Clip
- Mediafly
- Pristine.io
- CMUNE
- NodeCraft
- Platzi
- Workshape.io

Installation of RethinkDB

The official packages of RethinkDB for Ubuntu Linux, OS X, CentOS, Debian and Windows are available. In addition, community-supported packages are available on rethinkdb.com for Arch Linux, OpenSUSE, Fedora, Linux Mint, Raspbian and Gentoo.

Official RethinkDB client drivers are available for JavaScript, Ruby, Python and Java, while community-supported drivers are available for C#, Clojure, Delphi, Go, Lua, PHP, Swift, R, Common Lisp, Elixir, Haskell, Nim, C++, Dart, Erlang, JS Neumino, Perl and Rust. Drivers with limited features are also available for Objective-C and Scala.

Installing RethinkDB on Windows

The 64-bit binaries of RethinkDB for Windows 7 and its versions are available on the official URL of RethinkDB <https://www.rethinkdb.com/docs/install/windows/>. To install RethinkDB on Windows, the 64-bit flavour is required.

A ZIP archive is downloaded and unpacked in any

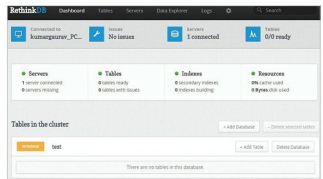


Figure 3: Dashboard and control panel of RethinkDB